

Sample Research Projects

Research in Information Security (Monsoon 2026)

1) Project Title:- Certificate-based proxy signature for IoT environment

Reference: Girraj Kumar Verma, B.B. Singh, Neeraj Kumar, Mohammad S. Obaidat, Debiao He, Harendra Singh, An efficient and provable certificate-based proxy signature scheme for IIoT environment, *Information Sciences*, Volume 518, 2020, Pages 142-156, <https://doi.org/10.1016/j.ins.2020.01.006>.
(Link:- <https://www.sciencedirect.com/science/article/pii/S0020025520300074>)

2) Project Title:- Proxy Re-Encryption to Secure Data Sharing in the Internet of Things Based on Blockchain

Reference: K. O. -B. O. Agyekum, Q. Xia, E. B. Sifah, C. N. A. Cobblah, H. Xia and J. Gao, "A Proxy Re-Encryption Approach to Secure Data Sharing in the Internet of Things Based on Blockchain," in *IEEE Systems Journal*, vol. 16, no. 1, pp. 1685-1696, March 2022, doi: 10.1109/JSYST.2021.3076759.
(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9442931>)

3) Project Title:- Hierarchical Multi Blockchain for Fine Grained Access to Medical Data

Reference: V. Malamas, P. Kotzanikolaou, T. K. Dasaklis and M. Burmester, "A Hierarchical Multi Blockchain for Fine Grained Access to Medical Data," in *IEEE Access*, vol. 8, pp. 134393-134412, 2020, doi: 10.1109/ACCESS.2020.3011201.
(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9146294>)

4) Project Title:- Identity-Based Decentralized Multi-Signature in Internet of Things

Reference: H. Liu, D. Han, M. Cui, K. -C. Li, A. Souri and M. Shojafar, "IdenMultiSig: Identity-Based Decentralized Multi-Signature in Internet of Things," in *IEEE Transactions on Computational Social Systems*, vol. 10, no. 4, pp. 1711-1721, Aug. 2023, doi: 10.1109/TCSS.2022.3232173.
(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10008196>)

5) Project Title:- Attribute-Based Hierarchical Access Control

Reference: M. Xiao, H. Li, Q. Huang, S. Yu and W. Susilo, "Attribute-Based Hierarchical Access Control With Extendable Policy," in *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 1868-1883, 2022, doi: 10.1109/TIFS.2022.3173412.
(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9770846&tag=1>)

6) Project Title:- Certificate-Less Aggregate Signature Scheme for Vehicular Ad Hoc Networks Enhanced Smart Grid System

Reference: Vallent, T.F.; Hanyurwimfura, D.; Mikeka, C. Efficient Certificate-Less Aggregate Signature Scheme with Conditional Privacy-Preservation for Vehicular Ad Hoc Networks Enhanced Smart Grid System. *Sensors* 2021, 21, 2900. <https://doi.org/10.3390/s21092900> (Link:- <https://www.mdpi.com/1424-8220/21/9/2900>)

7) Project Title:- Certificate-Based Aggregate Signature Scheme for Industrial Internet of Things

Reference: Z. Qiao et al., "An Efficient Certificate-Based Aggregate Signature Scheme With Provable Security for Industrial Internet of Things," in *IEEE Systems Journal*, vol. 17, no. 1, pp. 72-82, March 2023, doi: 10.1109/JSYST.2022.3188012. (Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9833446>)

8) Project Title:- Identity-Based Multi-Signature Scheme for Internet of Vehicles

Reference: V. Srivastava, S. K. Debnath, B. Bera, A. K. Das, Y. Park and P. Lorenz, "Blockchain-Envisioned Provably Secure Multivariate Identity-Based Multi-Signature Scheme for Internet of Vehicles Environment," in *IEEE Transactions on Vehicular Technology*, vol. 71, no. 9, pp. 9853-9867, Sept. 2022, doi: 10.1109/TVT.2022.3176755. (Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9779932>)

9) Project Title:- Design Secure Data Transmission in IoT

Reference:- Harbi, Yasmine, et al. "Enhanced authentication and key management scheme for securing data transmission in the internet of things." *Ad Hoc Networks* 94 (2019): 101948. (Link:- <https://www.sciencedirect.com/science/article/pii/S1570870519302689>)

10) Project Title:- Designing a Machine Learning Enabled Secure Cloud

Reference:- Ahmad, Shahnawaz, et al. "Machine learning-based intelligent security framework for secure cloud key management." *Cluster Computing* (2024): 1-27. (Link:- <https://link.springer.com/article/10.1007/s10586-024-04288-8>)

11) Project Title:- Design Secure Key Management Authentication in IoV

Reference:- H. Tan, et al. "Secure and Efficient Authenticated Key Management Scheme for UAV-Assisted Infrastructure-Less IoVs," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 6, pp. 6389-6400, June 2023. (Link:- <https://ieeexplore.ieee.org/abstract/document/10074983/authors#authors>)

12) Project Title:- Blockchain-Enabled Security Framework Against Ransomware Attacks for Smart Healthcare

Reference: Mohammad Wazid, et al. "BSFR-SH: Blockchain-Enabled Security Framework Against Ransomware Attacks for Smart Healthcare," in IEEE Transactions on Consumer Electronics, Vol. 69, No. 1, pp. 18-28, February 2023, DOI: 10.1109/TCE.2022.3208795. (2021 SCI Impact Factor: 4.414)

(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9899708>)

13) Project Title:- Post-Quantum Public-Key Searchable Encryption Scheme on Blockchain for E-Healthcare Scenarios

Reference: Xu, Gang, Xu, Shiyuan, Cao, Yibo, Yun, Fan, Cui, Yu, Yu, Yiying, Xiao, Ke, PPSEB: A Postquantum Public-Key Searchable Encryption Scheme on Blockchain for E-Healthcare Scenarios, Security and Communication Networks, 2022, 3368819, 13 pages, 2022.

<https://doi.org/10.1155/2022/3368819>

(Link:- <https://onlinelibrary.wiley.com/doi/epdf/10.1155/2022/3368819>)

14) Project Title:- Quantum-Resistant Password-Authenticated Searchable Encryption for Cloud Storage

Reference: J. Jiang and D. Wang, "QPASE: Quantum-Resistant Password-Authenticated Searchable Encryption for Cloud Storage," in IEEE Transactions on Information Forensics and Security, vol. 19, pp. 4231-4246, 2024, doi: 10.1109/TIFS.2024.3372804.

(Link:- <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10458707>)

15) Project Title:- Design of AI/ML-enabled Big data analytics for IoT environment

Reference: Singh, S. K., et al. Blockiotintelligence: A blockchain-enabled intelligent IoT architecture with artificial intelligence. Future Generation Computer Systems, 2020, 110, 721-743

(Link:- <https://www.sciencedirect.com/science/article/pii/S0167739X19316474>)

16) Project Title:- Anomaly Detection Using Machine Learning for Advanced Persistent Threats (APT)

Reference: Saini N, et al. A hybrid ensemble machine learning model for detecting APT attacks based on network behavior anomaly detection. *Concurrency Computat Pract Exper.* 2023; 35(28).

(Link:- <https://onlinelibrary.wiley.com/doi/10.1002/cpe.7865>)

17) Project Title:- Malware Detection Using Machine Learning in IoT Environment

Reference: Sumit Pundir, et al. 2021. MADP-IIME: malware attack detection protocol in IoT-enabled industrial multimedia environment using machine learning approach. *Multimedia Syst.* 29, 3 (Jun 2023), 1785–1797.

(Link:- <https://link.springer.com/article/10.1007/s00530-020-00743-9>)

28) Project Title:- Design Edge-Computing Enabled Data Collection and Finding Scheme for IoV

Reference: H. Sikarwar and D. Das, "SecEdge: Secure Edge-Computing-Based Hybrid Approach for Data Collection and Searching in IoV," in *IEEE Transactions on Network and Service Management*, vol. 21, no. 1, pp. 1213-1225, Feb. 2024, doi: 10.1109/TNSM.2023.3299264. (Link:- <https://ieeexplore.ieee.org/document/10196053>)

19) Project Title:- Multi-Party Computation based Federated Learning

Reference: R. Kanagavelu *et al.*, "Two-Phase Multi-Party Computation Enabled Privacy-Preserving Federated Learning," *2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID)*, Melbourne, VIC, Australia, 2020, pp. 410-419, doi: 10.1109/CCGrid49817.2020.00-52 (Link:- <https://ieeexplore.ieee.org/abstract/document/9139658>)

20) Project Title:- Quantum Teleportation-based Privacy Enabled Signature Scheme

Reference: P. Kumar *et al.*, "A Secure and Privacy-Preserving Signature Protocol Using Quantum Teleportation in Metaverse Environment," in *IEEE Access*, vol. 12, pp. 96718-96728, 2024, doi: 10.1109/ACCESS.2024.3427268. (Link:- <https://ieeexplore.ieee.org/document/10596298>)

21) Project Title:- Quantum Key Distribution in Trusted QKD Network

Reference: Chen, Li-Quan, et al. "A quantum key distribution routing scheme for hybrid-trusted QKD network system." *Quantum Information Processing* 22.1 (2023). (Link:- <https://link.springer.com/article/10.1007/s10586-024-04288-8>)

22) Project Title:- Isogeny-based Post-Quantum Proxy Signature for Internet of Things

Reference: Kunal Dey, Somnath Kumar, Vikas Srivastava, Sumit Kumar Debnath, Ashok Kumar Das, and Shehzad Ashraf Chaudhry. "Isogeny-based Post-Quantum Proxy Signature for Internet of Things," in *Scientific Reports* (Nature), July 2026. (2025 SCI Impact Factor: 4.9) [<http://www.nature.com/scientificreports>] [Link: <https://www.nature.com/articles/s41598-026-61779-x>]

23) Project Title:- Post-Quantum Secure Lattice-Based Authentication Scheme for IoT-Enabled Crop Recommender System

Reference: Snehal Jain, Abhishek Kumar Pandey, Ashok Kumar Das, Shantanu Pal, and Youngho Park. "Post-Quantum Secure Lattice-Based Authentication Scheme for IoT-Enabled Crop Recommender System," in *IEEE Internet of Things Journal*, June 2026, DOI: 10.1109/JIOT.2026.3709484. (2025 SCI Impact Factor: 8.7) [Link: <https://ieeexplore.ieee.org/document/11593814>]

24) Project Title:- Authenticated Key Agreement Scheme for Securing IoMT-Based Consumer Smart Healthcare Using Reinforcement Learning-as-a-Service

Reference: Abhishek Kumar Pandey, Ashok Kumar Das, Shantanu Pal, Bharat K. Bhargava, and Chengzu Dong. "Authenticated Key Agreement Scheme for Securing IoMT-Based Consumer Smart Healthcare Using Reinforcement Learning-as-a-Service," in *IEEE Transactions on Consumer Electronics*, June 2026, DOI: 10.1109/TCE.2026.3709614. (2025 SCI Impact Factor: 9.9) [Link: <https://ieeexplore.ieee.org/document/11593858>]

25) Project Title:- Multivariate Post-Quantum and Hash-Based Cryptographic Framework for Real-Time Numerical Thyroid Biomarker Anomaly Detection in IoMT

Reference: Jing Yang, Mohammed Okmi, Sunil Prajapat, Lip Yee Por, Ashok Kumar Das, Joel J. P. C. Rodrigues, and Xiaojie Ji. "NeuroThyroid-PQC: A Multivariate Post-Quantum and Hash-Based Cryptographic Framework for Real-Time Numerical Thyroid Biomarker Anomaly Detection in IoMT," in *IEEE Internet of Things Journal*, May 2026, DOI: 10.1109/JIOT.2026.3699804. (2025 SCI Impact Factor: 8.7) [Link: <https://ieeexplore.ieee.org/document/11550136>]